Executive Summary Report: Strategic AI-Enabled Transformation of KazMunayGas

Comparative Strategic Analysis & Enterprise Implementation Plan

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Company A: Shell **Company B:** KazMunayGas (KMG)

1. Executive Position & Core Board Recommendation

KazMunayGas (KMG) must deploy artificial intelligence strictly as an operational driver for predictive asset maintenance, subsoil reservoir management, and industrial safety, while explicitly rejecting Shell's capital strategy of unhedged retail price floating and wholesale downstream asset divestment.

As documented in the corporate disclosures of Joint Stock Company National Company KazMunayGas, the corporation operates under a sovereign dual mandate. Controlling 67.42% of KMG's equity through Samruk-Kazyna JSC, alongside 20.00% held by the Ministry of Finance and 9.58% by the National Bank of the Republic of Kazakhstan, the Kazakh state requires KMG to generate commercial returns while guaranteeing domestic energy security, maintaining affordable fuel supplies, and preserving regional employment for over 52,000 workers across West Kazakhstan.

Core Board Directive: The primary Board recommendation is to accelerate enterprise-wide scaling of predictive failure analytics across domestic refining assets specifically the Atyrau Refinery (ANPZ) and Pavlodar Oil Chemistry Refinery (PNHZ) and expand the ABAI information system across upstream operations. To protect institutional stability, all operational AI applications must function under a Human-in-the-Loop governance regime where named engineers retain exclusive legal and fiduciary accountability.

2. Macro-Business Context & Financial Resilience: KMG vs. Shell plc

A comparative financial and operational analysis establishes fundamental structural divergence between global energy major Shell plc and national champion KazMunayGas. According to audited consolidated financial statements, KMG demonstrated strong operational resilience in 2025, generating consolidated revenue of 9,371 billion KZT ($17.98 billion), earnings before interest, taxes, depreciation, and amortization (EBITDA) of 2,393 billion KZT ($4.59 billion), and a net profit of 1,072 billion KZT ($2.00 billion). KMG maintained an exceptionally conservative balance sheet with a Net Debt-to-EBITDA leverage ratio of 0.16x, supported by 1,848 billion KZT in liquid cash reserves against 2,223 billion KZT in total gross debt.

In physical volume, KMG produced 26.211 million tonnes of crude oil and gas condensate, representing 26% of Kazakhstan's total national hydrocarbon output. In transportation, KMG conveyed 83 million tonnes of crude accounting for 55% of national pipeline and maritime transit and refined 21.0 million tonnes of crude across domestic and European facilities, supplying 80% of Kazakhstan's domestic fuel market.

In contrast, comparative benchmark Shell plc operates in open international capital markets focused on shareholder cash returns. In 2025, Shell produced 2.800 million barrels of oil equivalent per day,

generating $42.9 billion in Cash Flow from Operations and $26.1 billion in Free Cash Flow. Shell allocated $22.4 billion representing 52% of its operating cash flow to shareholder distributions through $8.5 billion in dividends and $13.9 billion in share buybacks, while divesting carbon-intensive legacy assets to achieve a 36% reduction in Scope 1 and Scope 2 operational emissions.

Table 1 provides a side-by-side financial and operational comparison highlighting the contrasting mandates and structural profiles of both enterprises:

Financial & Operational Metric	KazMunayGas (Company B)	Shell plc (Company A)	Strategic Significance
Primary Mandate	National energy security & state development	Maximizing shareholder value & FCF per share	KMG prioritizes supply continuity over retail margin optimization.
Consolidated Revenue	9,371 Billion KZT ($17.98B)	$316.4 Billion	Shell operates at global scale; KMG acts as national economic anchor.
EBITDA / Operating Profit	2,393 Billion KZT ($4.59B)	$68.5 Billion	Both enterprises maintain high operational cash flow generation.
Net Debt / EBITDA Leverage	0.16x (Net Debt: 375B KZT)	~0.63x (Net Debt: $43.5B)	KMG maintains a conservative balance sheet buffer against volatility.
Hydrocarbon Production	26.211M tonnes (~544 kbopd)	2.800M boe/day	KMG controls 26% of Kazakhstan national output; Shell manages global portfolio.
Refining & Market Share	21.0M tonnes (80% domestic share)	Integrated global refining & LNG	KMG acts as primary supplier for domestic transport and agriculture.
Capital Allocation Focus	Sustaining infrastructure & low leverage	Share buybacks ($13.9B) & dividends ($8.5B)	Shell returns cash to equity; KMG reinvests in national infrastructure.

3. Strategic Transferability Matrix & Policy Alignment

Directly copying Shell's corporate practices would violate Kazakhstan's legal regulations and compromise domestic socio-economic stability. Managerial practices and digital technologies evaluated across the five-level transferability framework yield clear strategic verdicts:

- **ADOPT: Enterprise Predictive Maintenance & Safety Vision Systems:** KMG should fully adopt Shell's practice of enterprise-wide predictive failure analytics and computer-vision safety monitoring. Extending the proprietary TUMAR safety vision system across 60 well workover

crews and scaling predictive vibration analytics across domestic refineries eliminates unplanned outages and protects frontline workers.

- **ADAPT: Decarbonization Framework & Hybrid Power Generation:** Shell's Net-Zero framework must be adapted to Kazakhstan's fossil-reliant economy. Rather than divesting core production, KMG should focus on Scope 1 and Scope 2 methane abatement under the OGMP 2.0 Gold Standard, maintain an associated petroleum gas utilization rate of 98.8%, and construct hybrid renewable facilities such as the 247 MW solar/wind/gas power plant in Zhanaozen executed in partnership with Eni.
- **PILOT: Advanced EOR & Slim Hole Subsoil Technology:** Shell's digital twin subsoil modeling and enhanced oil recovery techniques should be piloted across declining mature fields, including Embamunaygas and Ozenmunaygas. Scaling Slim Hole drilling technology at the Uzen field has already demonstrated a 30% reduction in well construction CAPEX, yielding 4.6 billion KZT in direct capital savings.
- **DELAY: Capital Allocation to Green Hydrogen & Retail EV Networks:** KMG should delay large-scale capital expenditure in green hydrogen electrolysers and retail electric vehicle charging infrastructure until regional EV adoption matures, restricting current involvement to preliminary research studies.
- **REJECT: Dynamic Fuel Price Floating & Asset Divestment:** Shell's strategy of floating retail fuel prices and divesting lower-margin refining assets must be explicitly rejected. Under Order No. 171 of the Minister of Energy of the Republic of Kazakhstan, domestic retail prices for AI-92 petrol and diesel are subject to strict statutory price caps. Implementing dynamic surge pricing or selling domestic refineries would trigger severe domestic inflation, transport disruption, and regional social distress.

4. Human-AI Operating Model & Task Allocation Architecture

To operationalize AI technology safely, KMG assigns operational responsibilities across the four functions of management Planning, Organizing, Leading, and Controlling using a strict four-category allocation framework:

- **Human Only:** Tasks requiring moral agency, socio-political empathy, and legal liability remain exclusively human. Reallocating workforce headcount in the Mangystau region and conducting executive performance reviews are classified as Human Only due to binding labor obligations under the Labor Code of the Republic of Kazakhstan and KMG's social mandate across 52,000 workers.
- **Human + AI:** High-stakes strategic decisions utilize AI scenario modeling while retaining human authority. Allocating annual capital expenditure across regional subsidiaries and planning decarbonization investments combine algorithmic financial modeling with executive sign-off from the Chief Financial Officer.
- **AI Executes + Human Review:** High-volume diagnostic tasks delegate data processing to algorithms under mandatory human verification. Running predictive failure algorithms on refinery pumps and adjusting waterflooding injection rates across 1,660 cells on the ABAI platform operate

under this mode. A certified Chief Mechanical Engineer or Chief Reservoir Engineer must physically review diagnostic telemetry before approving maintenance work orders in SAP ERP.

- **Automated:** Low-risk, high-volume tasks governed by rigid rules operate end-to-end automatically. Real-time SCADA sensor telemetry collection and routine environmental compliance reporting operate automatically, supported by a manual logbook fallback protocol.

Table 2 outlines the operational task allocation across POLC functions:

POLC Function	Operational Task	Allocation Category	Accountable Role
Planning	Annual CAPEX Allocation Across Assets	**Human + AI**	CFO / Executive Board
Planning	Regional Fuel Demand Forecasting	**AI Executes + Human Review**	Head of Marketing
Organizing	Mangystau Workforce Restructuring	**Human Only**	HR Director
Organizing	Routine Sensor Inspection Scheduling	**Automated**	Maintenance Lead
Leading	Executive & Employee Performance Review	**Human Only**	General Director
Leading	Workover Safety Compliance (TUMAR)	**AI Executes + Human Review**	Chief HSE Officer
Controlling	Refinery Predictive Failure Monitoring	**AI Executes + Human Review**	Chief Mechanical Engineer
Controlling	ABAI Waterflooding Rate Optimization	**AI Executes + Human Review**	Chief Reservoir Engineer

Operational Control Safeguards

To ensure operational resilience, KMG enforces three strict operational control safeguards:

1. Human Review Point: A certified Chief Mechanical Engineer must independently audit AI diagnostic telemetry and physically sign off before any refinery processing unit is shut down.

2. Escalation Path: When an AI anomaly flag occurs, the Senior Maintenance Technician is alerted within 15 minutes, the Chief Mechanical Engineer within 1 hour, and the General Director within 4 hours if estimated repair time exceeds 24 hours.

3. Manual Fallback Protocol: If IoT sensor networks or AI predictive servers experience power loss or cyber outage, plant operators immediately revert to manual hourly operator logbooks, analog pressure gauge walks, and standard calendar maintenance schedules.

5. Operational Process Engineering & Supply Chain Optimization

At the Atyrau Refinery, legacy equipment maintenance suffered from an operational bottleneck in manual trend analysis. Maintenance personnel spent 12 to 24 hours manually transcribing gauge readings into Excel spreadsheets, resulting in reactive maintenance performed after mechanical failure had already occurred. This friction contributed to an average baseline of 120 hours per year of unplanned refinery downtime.

Under the future-state process model, continuous 1 Hz IoT sensor telemetry streams directly into the ABAI platform's machine learning module. The algorithm detects micro-friction anomalies 14 days prior to potential failure, eliminating the manual analysis bottleneck. The operational process target is to achieve a 15% reduction in unplanned refinery downtime (reducing outages to under 102 hours per year) and cut maintenance OPEX by 10% by the end of 2027.

6. Data Governance Framework, Risk Register & Audit Safeguards

Industrial data processing requires a structured 5-Stage Data Refinement Workflow: Data Ingestion (1 Hz SCADA telemetry), Data Cleaning & Validation (automated quality score checks above 95%), AI Model Inference (predictive anomaly detection), Human Expert Review (Chief Mechanical Engineer sign-off), and Operational Execution (logging work orders in SAP ERP).

To protect critical infrastructure, KMG enforces a 7-Dimension Risk Register:

- **Privacy Risk:** Unauthorized exposure of employee records. Controlled via role-based access controls, data encryption over secure KMG VPN, and hosting at the Astana Data Center under Kazakh personal data protection statutes.
- **Cybersecurity Risk:** Ransomware attacks on refinery operational technology networks. Controlled by complete network air-gapping of SCADA control systems, mandatory two-factor authentication, and daily encrypted offsite backups.
- **Model Bias Risk:** ML algorithms trained on summer telemetry misinterpreting equipment wear at sub-zero winter temperatures (-30°C). Controlled via weekly model recalibration using 5-year seasonal climate datasets.
- **Reliability Risk:** AI model triggering false-positive shutdowns or missing critical bearing wear. Controlled by mandatory Human Review Points requiring physical vibrometer confirmation.
- **Accountability Gap Risk:** Engineers deflecting blame by claiming 'the AI model recommended it.' Controlled by explicit corporate policy designating AI as Decision Support System only, requiring personal engineer signatures on all work orders.
- **Escalation Failure Risk:** System failing to alert control rooms during rapid multi-pump vibration spikes. Controlled by dual redundant alarm triggers operating on independent physical circuits.

- **Manual Fallback Risk:** Operator skill degradation due to over-reliance on automation. Controlled via mandatory monthly manual control room drills.

Financial & Data Discrepancy Audit

Independent auditing of AI-generated analytical outputs against audited PwC financial statements exposed critical algorithmic hallucinations:

- **Debt-to-Equity Error:** AI initial output reported a Debt-to-Equity ratio of 0.92x and warned of insolvency. Human audit revealed AI incorrectly aggregated non-recourse joint-venture liabilities. The true verified D/E ratio is 0.24x.

- **Net Debt / EBITDA Error:** AI initially reported Net Debt/EBITDA of 0.45x. Human audit revealed AI failed to deduct 1,848 billion KZT in cash reserves from gross debt. The true verified Net Debt/EBITDA ratio is 0.16x.

- **Price Regulation Hallucination:** AI initially recommended dynamic surge pricing at Qazaq Oil retail stations during peak harvesting periods. Human audit rejected the claim as an illegal practice violating Order No. 171 of the Ministry of Energy RK.

7. 12-Month Implementation Roadmap & Capital Business Case

The strategic AI transformation plan executes across three structured phases over a 12-month timeline. Every initiative integrates all seven required operational parameters:

Phase 1: Foundation, Governance & Data Cleaning (Months 0–3)

- **Initiative 1.1 Scope:** Audit and clean SCADA sensor telemetry across Atyrau Refinery processing units.
- **Action:** Audit and standardize telemetry data ingestion streams.
- **Responsible Owner:** Chief Information Officer & Senior Reliability Engineer.
- **KPI Baseline & Target:** Elevate data quality score from 82% baseline to 98% by Month 3.
- **Dependencies:** Access to historical SAP ERP maintenance records.
- **Cost & Resources:** 45 million KZT for specialized data auditing software and local engineering hours.
- **Risks & Controls:** Incompatible legacy sensor formats; controlled via custom API translation scripts.
- **Decision Gate:** Proceed to Phase 2 only if data quality score exceeds 95% across all refinery units.

Phase 2: Pilot Deployment & Human-in-the-Loop Integration (Months 4–8)

- **Initiative 2.1 Scope:** Pilot ABAI predictive maintenance failure analytics at Atyrau Refinery catalytic cracking unit.
- **Action:** Integrate machine learning failure prediction algorithms with SCADA control interfaces.
- **Responsible Owner:** Chief Mechanical Engineer & Lead Data Scientist.

- **KPI Baseline & Target:** Achieve 14-day advance failure prediction lead time; reduce unplanned downtime by 10% from 120 hrs/yr baseline by Month 8.
- **Dependencies:** Completion of Phase 1 data cleaning audit.
- **Cost & Resources:** 120 million KZT for IoT vibration sensor retrofits and ABAI platform integration.
- **Risks & Controls:** High false-positive alarm rates; controlled by mandatory physical vibrometer inspection sign-off.
- **Decision Gate:** Scale to Pavlodar refinery only if false-positive rate remains below 5% during trial.

Phase 3: Enterprise Scaling, Training & Governance Assurance (Months 9–12)

- **Initiative 3.1 Scope:** Scale predictive failure analytics across Pavlodar refinery and expand ABAI waterflooding management across 1,660 injection cells.
- **Action:** Deploy validated predictive failure analytics across all domestic refining assets.
- **Responsible Owner:** Vice President of Refining Operations & KMG Engineering Director.
- **KPI Baseline & Target:** Achieve 15% reduction in total unplanned refinery downtime (<102 hrs/yr) and 10% reduction in maintenance OPEX by Month 12.
- **Dependencies:** Successful Phase 2 pilot sign-off by Chief Mechanical Engineer.
- **Cost & Resources:** 210 million KZT for full enterprise software scaling and technical staff training.
- **Risks & Controls:** Model drift under severe winter operating conditions (-30°C); controlled via weekly algorithm recalibration on 5-year climate datasets.
- **Decision Gate:** Maintain deployment if audited maintenance OPEX reduction achieves at least 8% annually.

Financial Business Case Summary

The total capital investment for the 12-month transformation is estimated at 375 million KZT ($720,000). Financial returns are driven by eliminating unplanned refinery outages (saving an estimated 500 million KZT per day of avoided emergency downtime) and optimizing reservoir waterflooding power usage (yielding an 8% to 12% reduction in artificial lift electricity OPEX). Based on audited operational proxy metrics, the project delivers full capital payback within 7.4 months of deployment, yielding a net present value of 1.42 billion KZT over a 3-year operating horizon while maintaining KMG's conservative 0.16x Net Debt/EBITDA leverage structure.